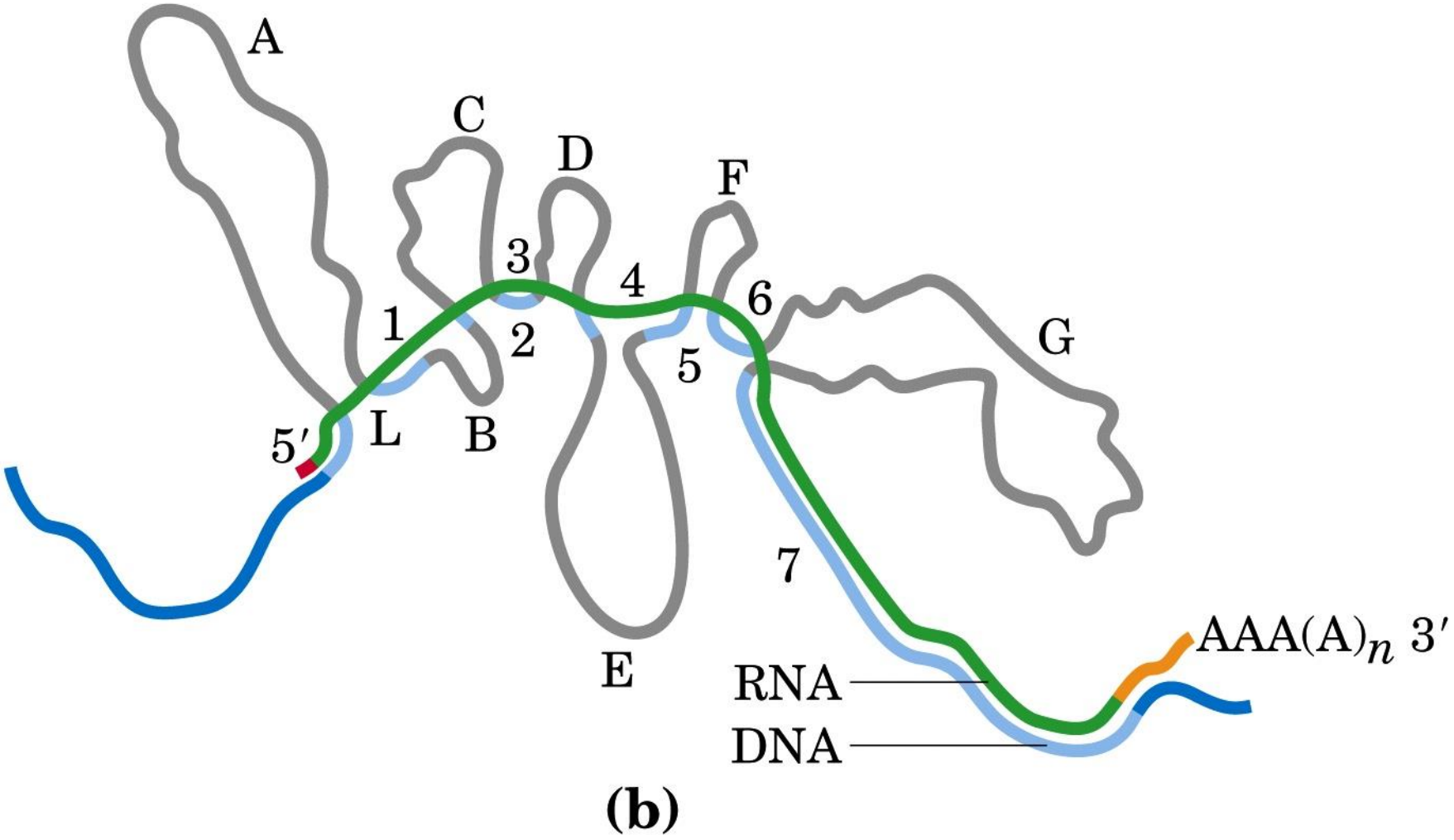
RNA Metabolism-Part2

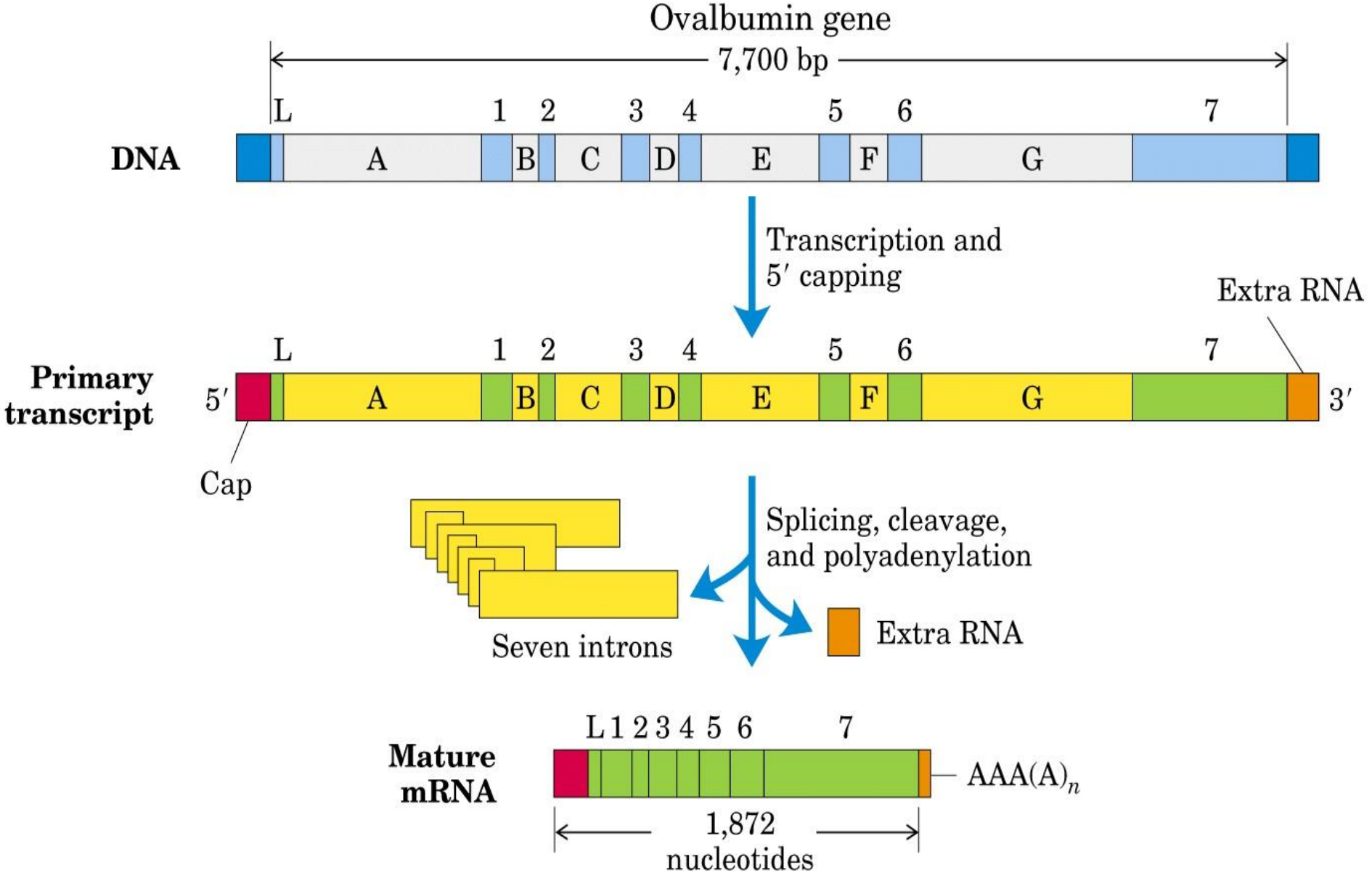
Post-transcriptional modifications

Dr. Rula Abdul Ghani

Modified RNA :



From Gene to mature mRNA



5’ cap of mRNA:

A- protects mRNA from ribonucleases

B- helps binding of mRNA to ribosomes.

7-methyl Guanosine linked to 5’ terminal residue of mRNA through 5’- 5’ triphosphate linkage.

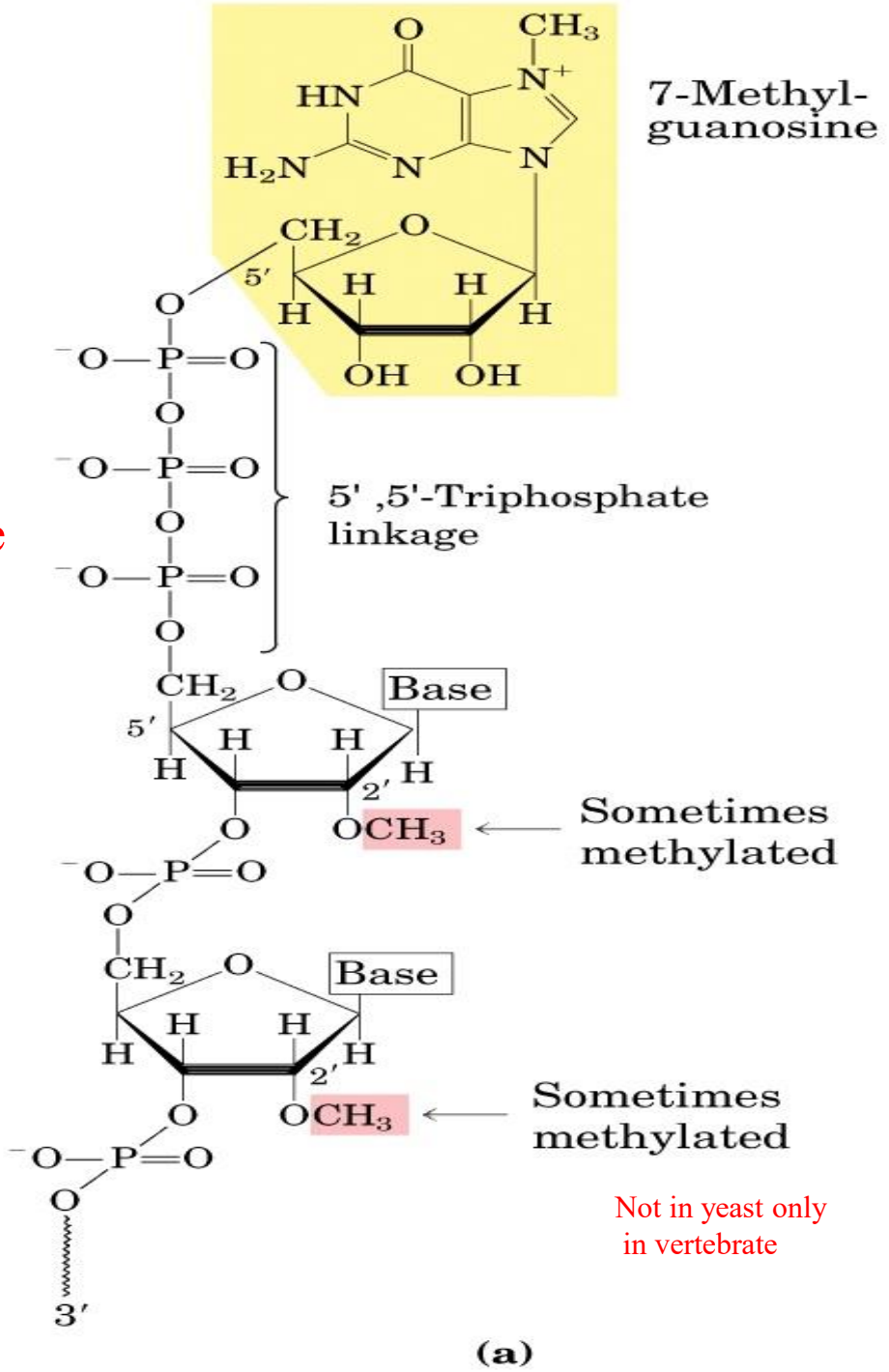
Formed from GTP with triphosphate at 5’ end.

After 20-30 nucleotides added (very early).

Guanine methylated at N-7 .

Additional methyl groups at C-2

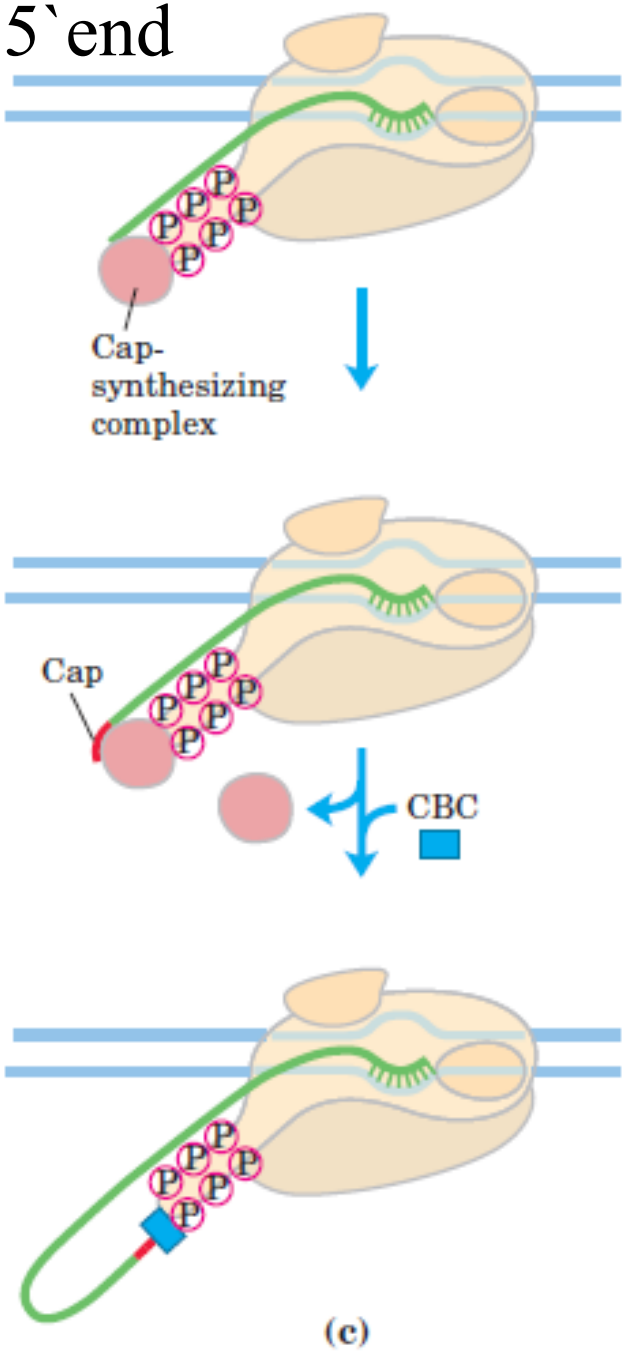
- Methyl donor = S-adenosyl Methionine.



Eukaryotic mRNA are capped at 5`end

Cap remains in association with Carboxy Terminal Domain CTD.

Cap binding complex CBC



Polyadenylation:

Poly A tail 80-250 A residues.

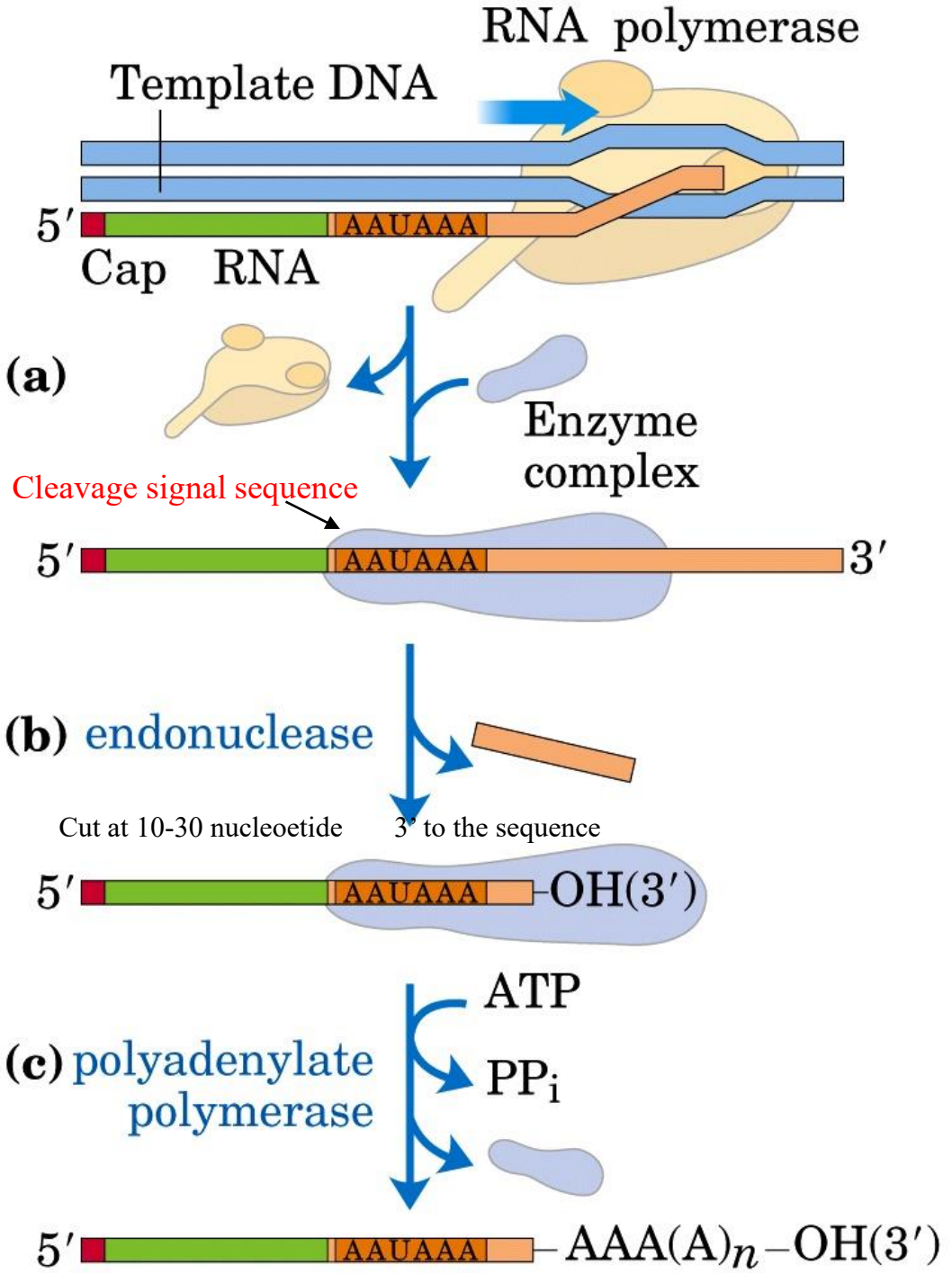
Protect RNA from enzymatic degradation

Polyadenylate polymerase.

RNA+ nATP →RNA-(AMP)n +nPPi

n =80-250

Requires:
cleaved RNA as a primer
No need for template.



RNA splicing :

4 groups/ classes:

Group I : guanosine cofactor. Nuclear, Mitochondrial, and genes that code for rRNA mRNA , tRNA

Group II: introns found in mitochondrial mRNA

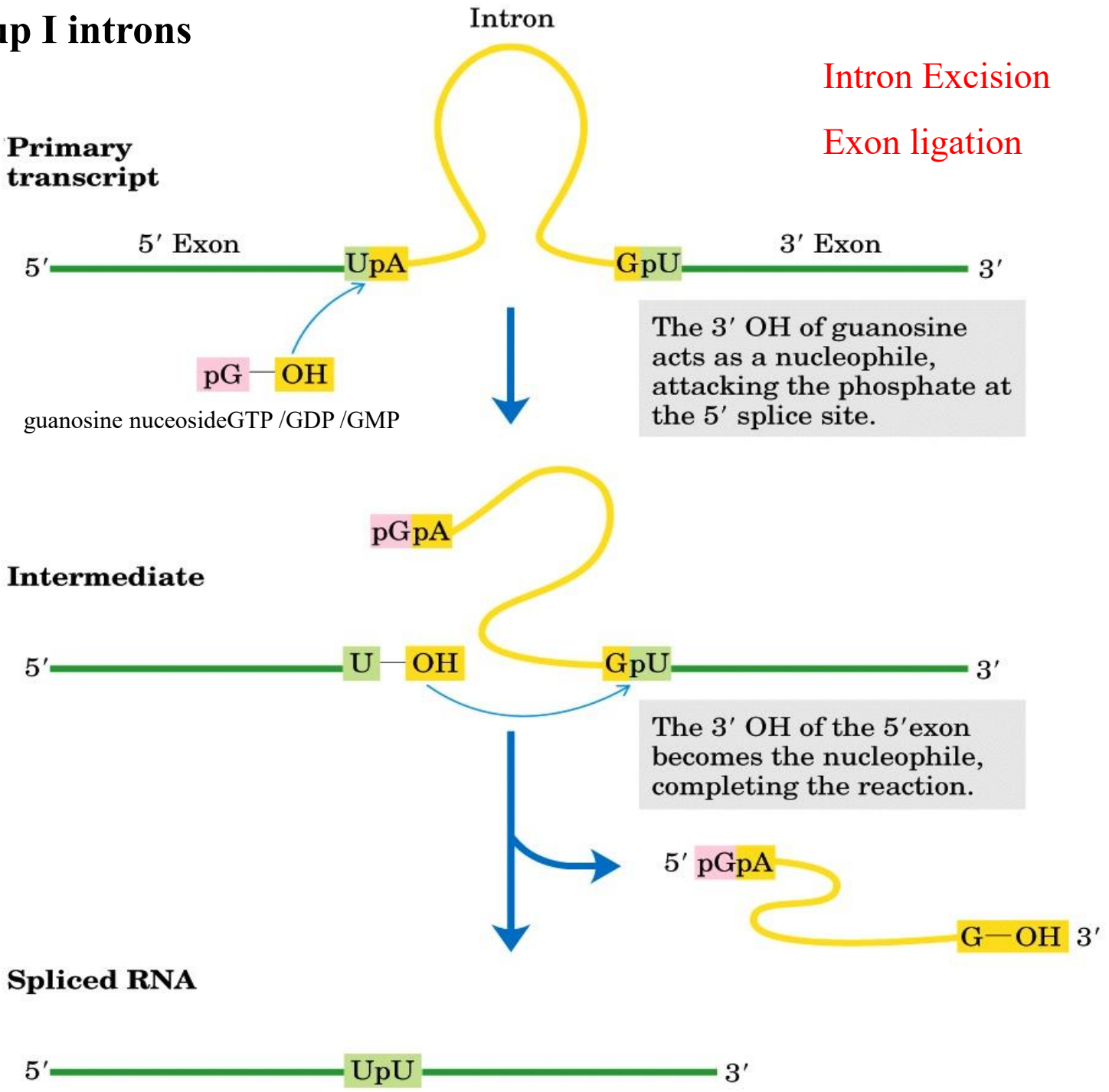
Group I + II : self splicing.

Group III : largest class splicesome = specialized RNA- protein complexes, consisting of a variety of proteins + small nuclear ribonucleoproteins (snRNPs) that recognize splice sites at the ends of RNA intron. Each snRNP contain one of U1 U2 U4 U5 U6 , releasing the intron & joining the two adjacent exons.

Nuclear mRNA.

Group four : tRNA, the only class spliced by protein enzymes. This needs ATP and an endonuclease.

RNA splicing: Group I introns



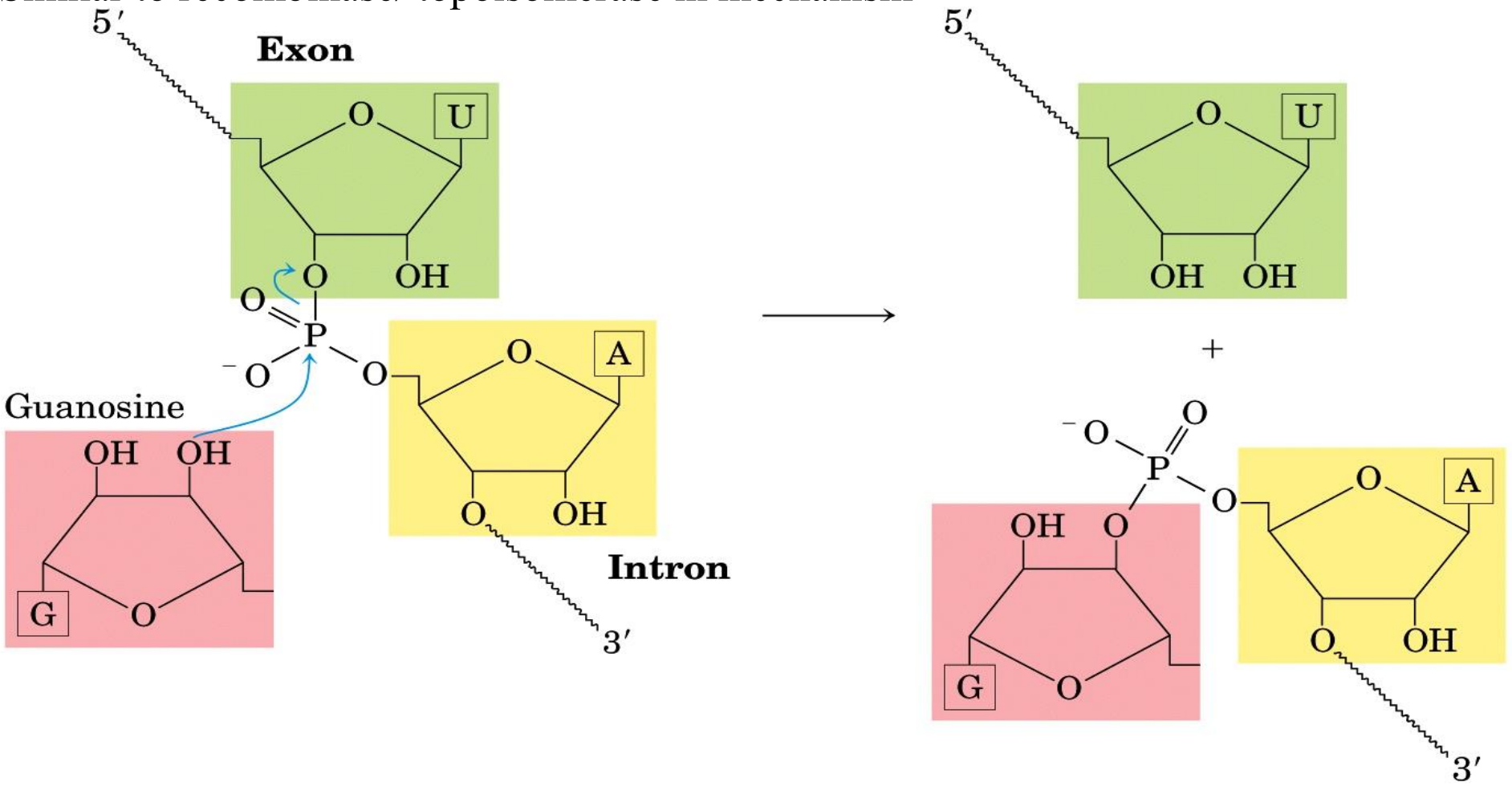
Transesterification reaction :

Self splicing: no protein enzymes required.

3’ OH of guanosine acts as a nucleophile.

New phosphodiester bond formed on expense of the old.

Similar to recombinase/ topoisomerase in mechanism



RNA Splicing : group II introns :

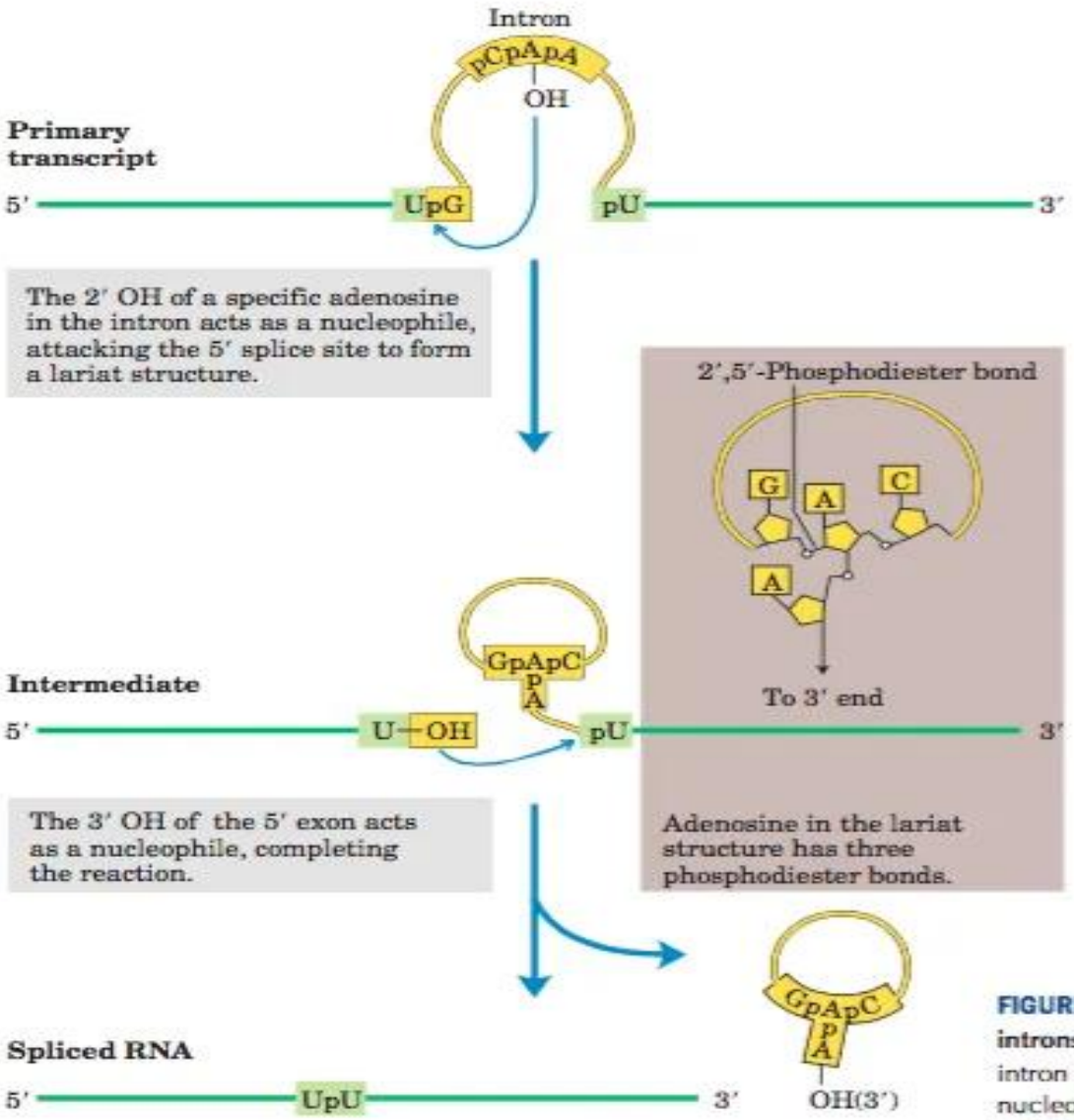
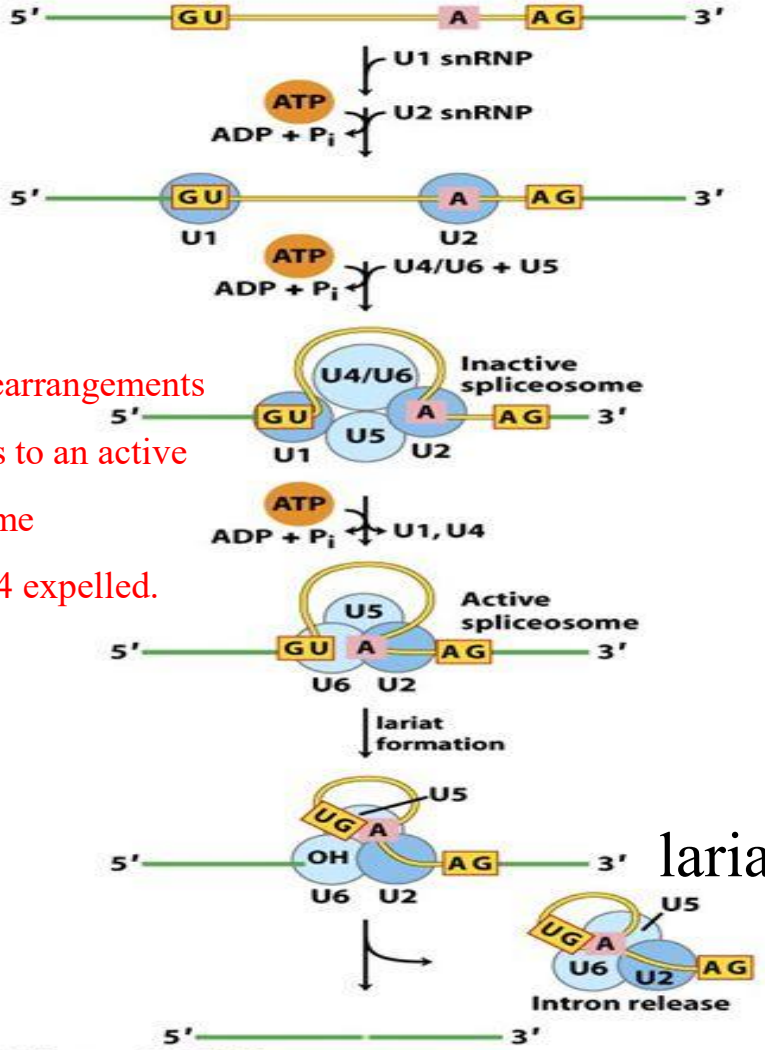


FIGURE 26–15 Splicing mechanism of group II introns. The chemistry is similar to that of group I intron splicing, except for the identity of the nucleophile in the first step and formation of a lariatlike intermediate, in which one branch is a 2′,5′-phosphodiester bond.

RNA processing : Group III introns

mRNA processing:
RNA splicing & editing
Mainly in eukaryotic organisms, highly conserved from yeast to human

Spliceosome is an RNA-protein complex; RNA proposed to be catalytic



Internal rearrangements covert this to an active spliceosome

U1 and U4 expelled.

Figure 26-17b
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Coordination of splicing with transcription

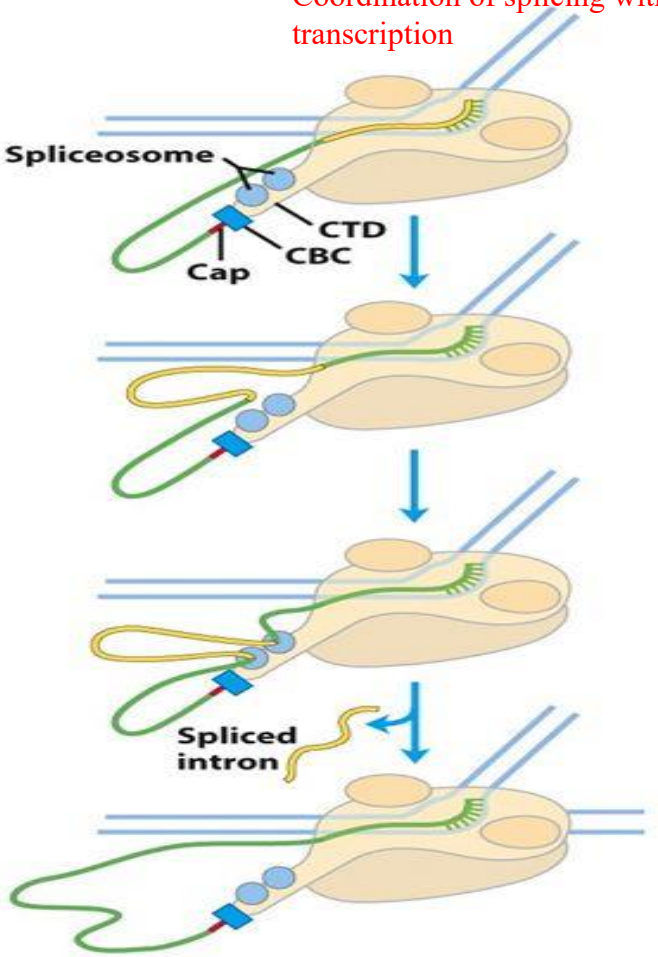
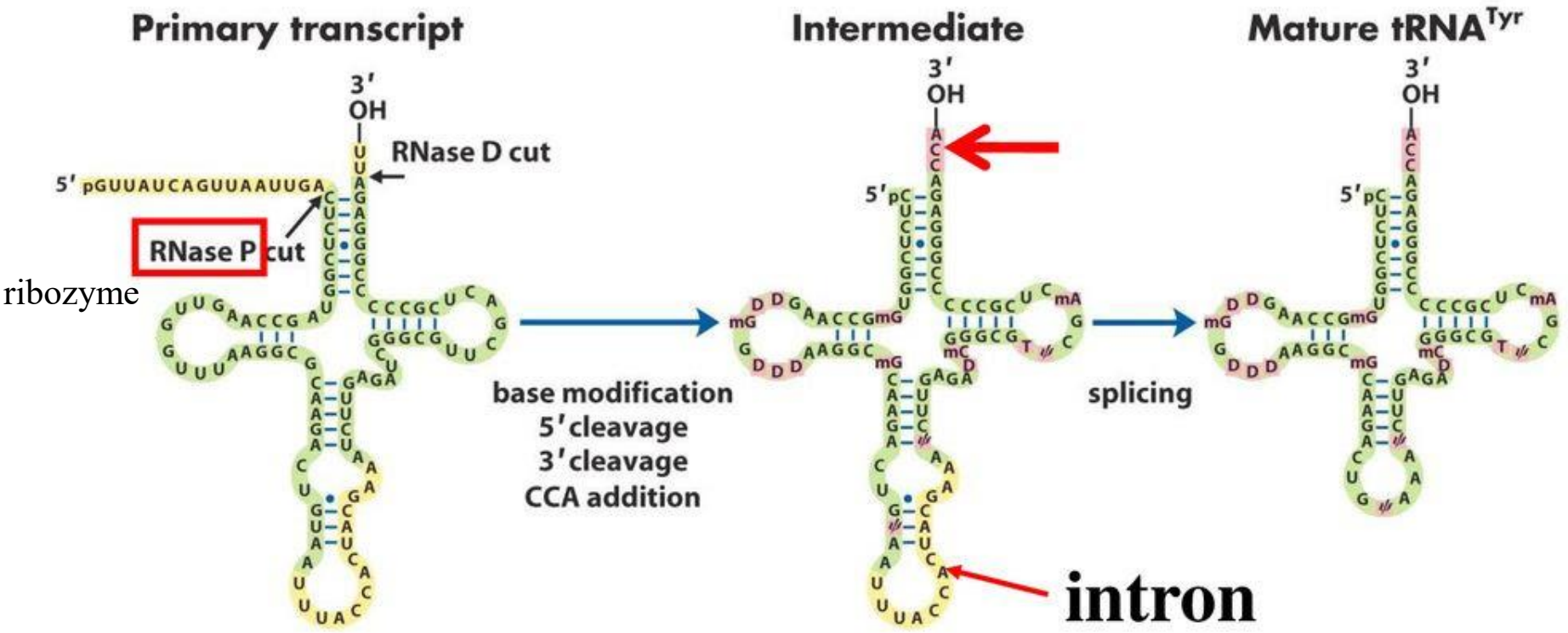


Figure 26-17c
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

RNA splicing: Group IV:

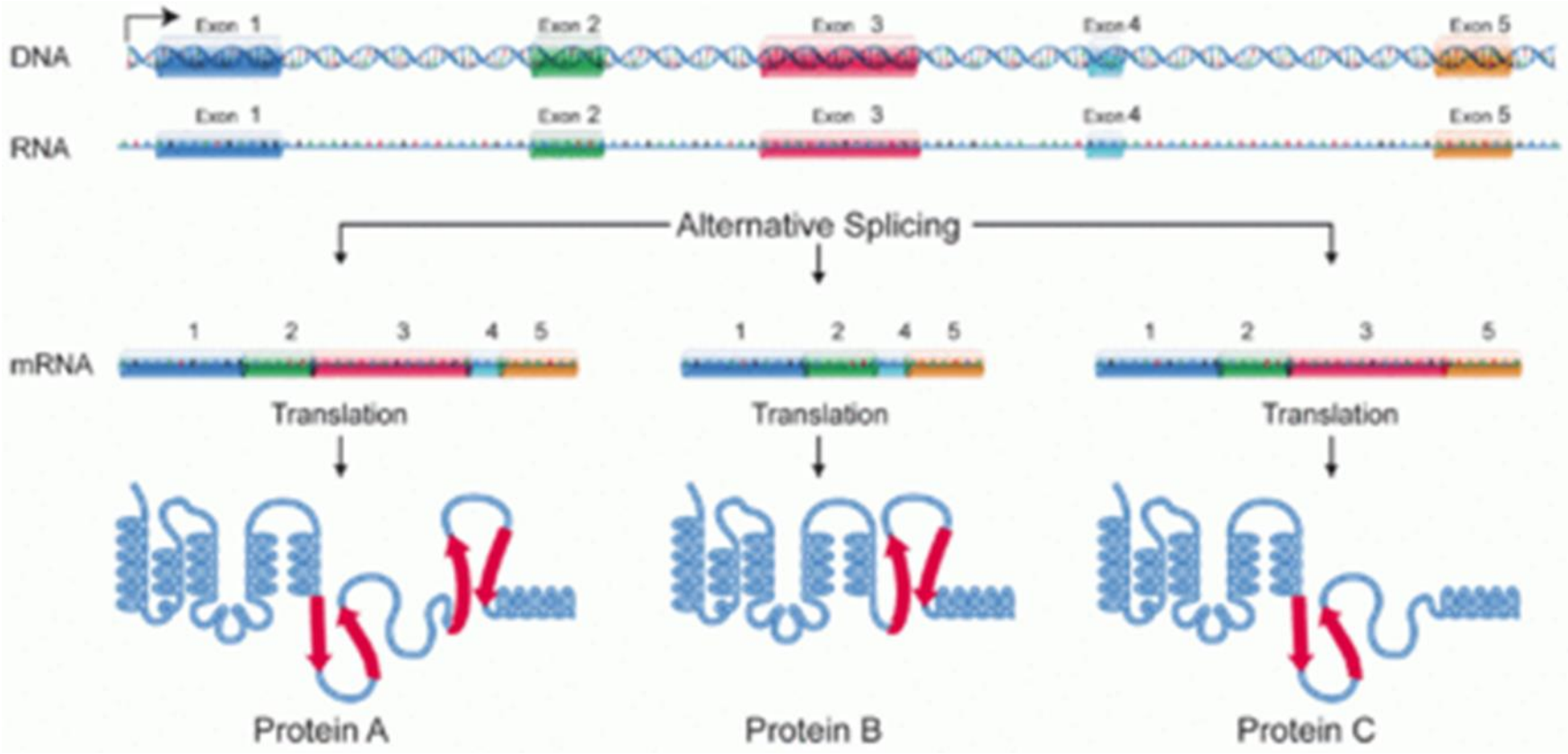


The processing of the primary tRNA transcripts include removal of the 5'and 3' ends, addition of the CCA sequence at the 3' end, modification of many bases, and splicing of introns (in eukaryotic cells).

Alternative splicing

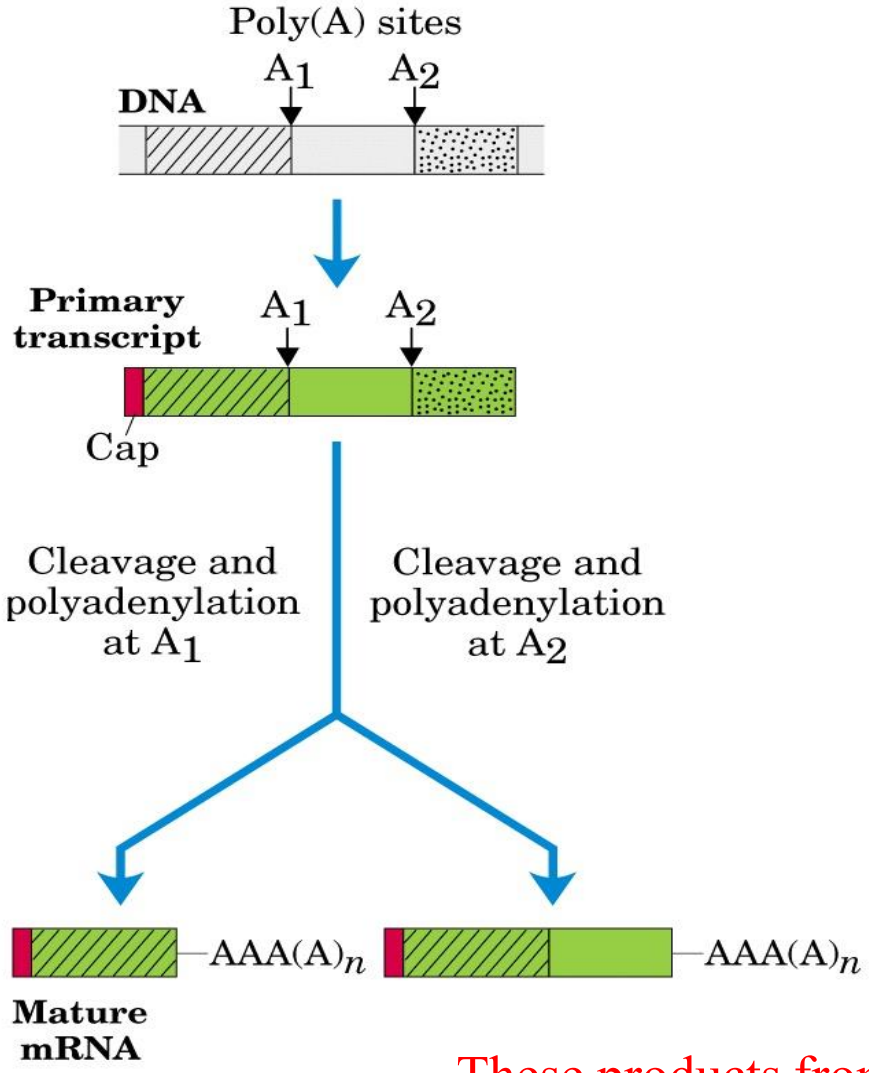
Why transcribe introns when spliced and not coded:

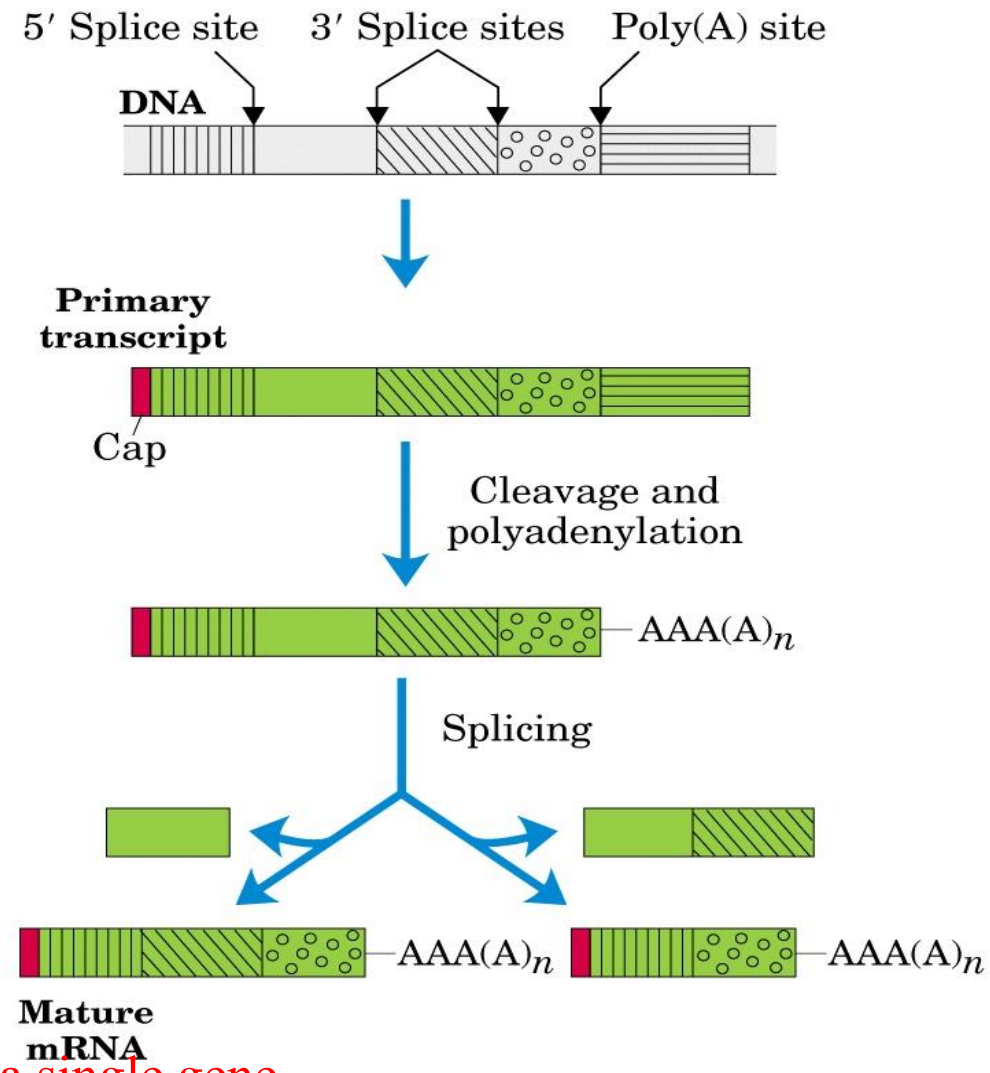
One gene→ several mature RNA transcripts→ several protein products.



Complex transcripts in different tissues (e.g brain vs. thyroid) :

a) More than one site for cleavage and polyadenylation.
b) Alternative splicing patterns.



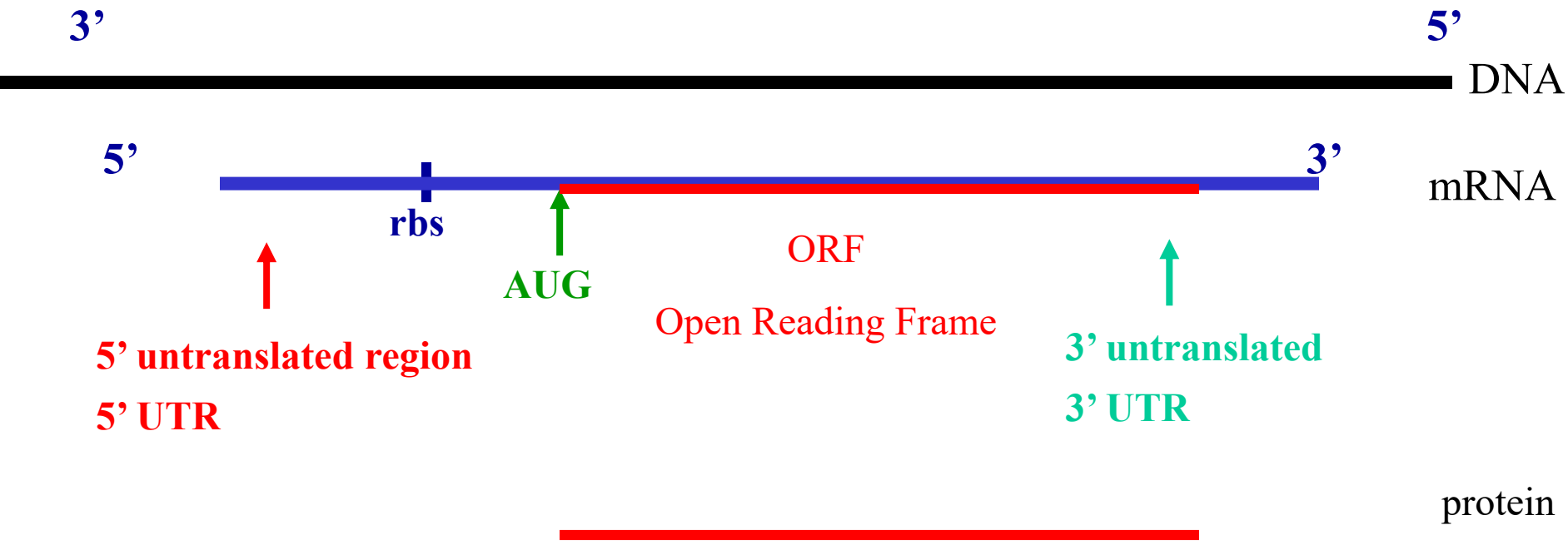


These products from a single gene

Transcript Structure

The 5'cap and polyA tail not translated into protein (nor are 5'UTR and 3' UTR).

- These modifications share several functions:
  - Facilitate the export of mRNA
  - Protect mRNA from degradation by hydrolytic enzymes
  - Help ribosomes attach to the 5' end



rbs: ribosome binding site

Gene Regulation

The process in which some genes turned on , others turned off.

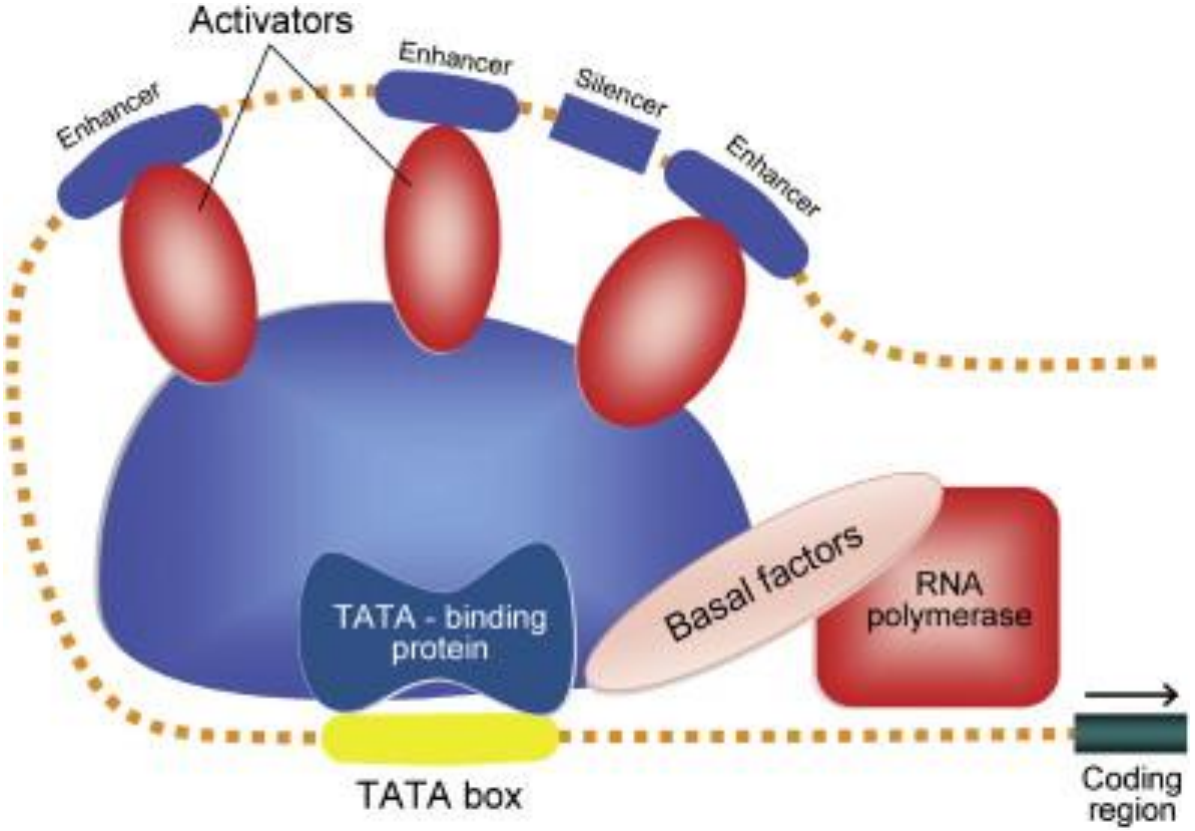
Inducible genes: only expressed when activators or cellular factors are present.

Housekeeping genes: are expressed constrictively since these are essential for the basic processes i.e replication and growth.

Regulatory sequences: Promoters , enhancers, activators silencer

Enhancers: DNA sequences that boost transcription and provide regulatory specificity (e.g. tissue specific expression)

- located *upstream* of the promoter (yeast: within 1kbp; humans: up to 10kbp away)



Promoter

Integration of retrovirus into host DNA.

Reverse transcriptase:

1st described in retroviruses as part of their life cycl

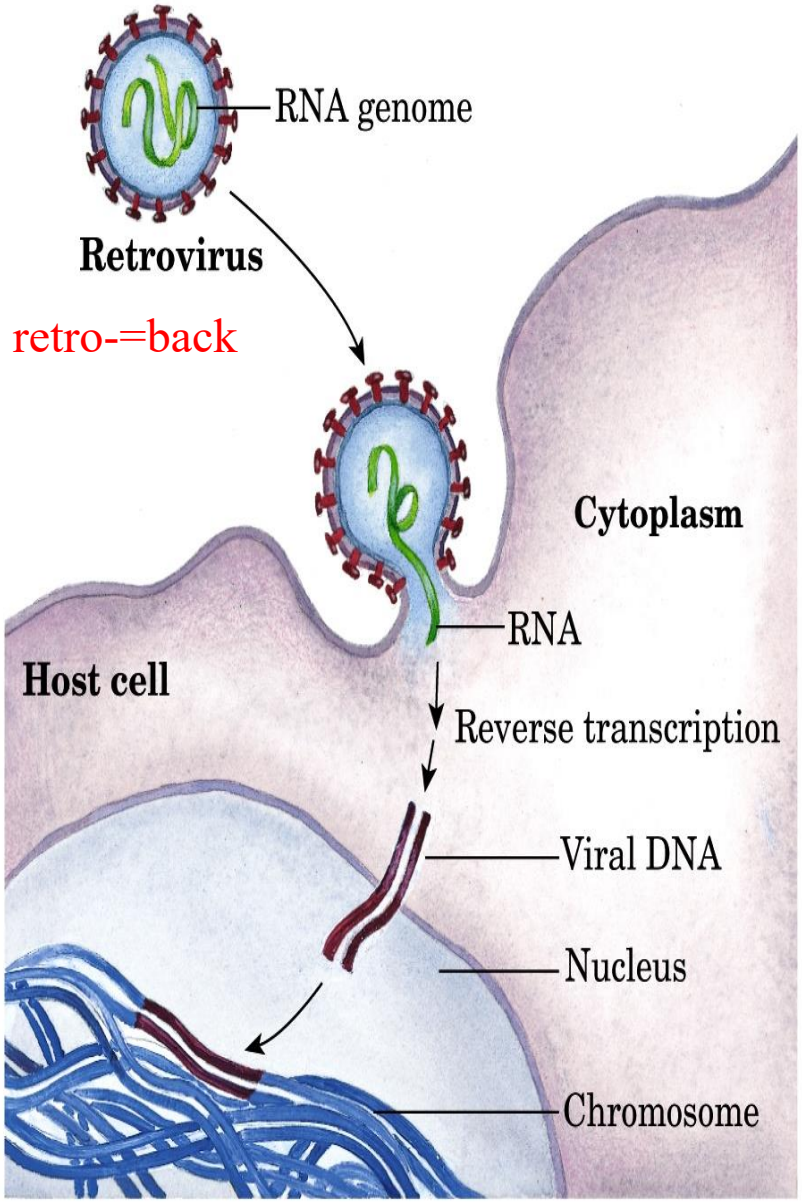
Requires a primer

No proofreading activity

(A PROBLEM IN RNA viruses)

Synthesis 5'→3'

DNA synthesized called cDNA



Inhibitor of reverse transcriptase:

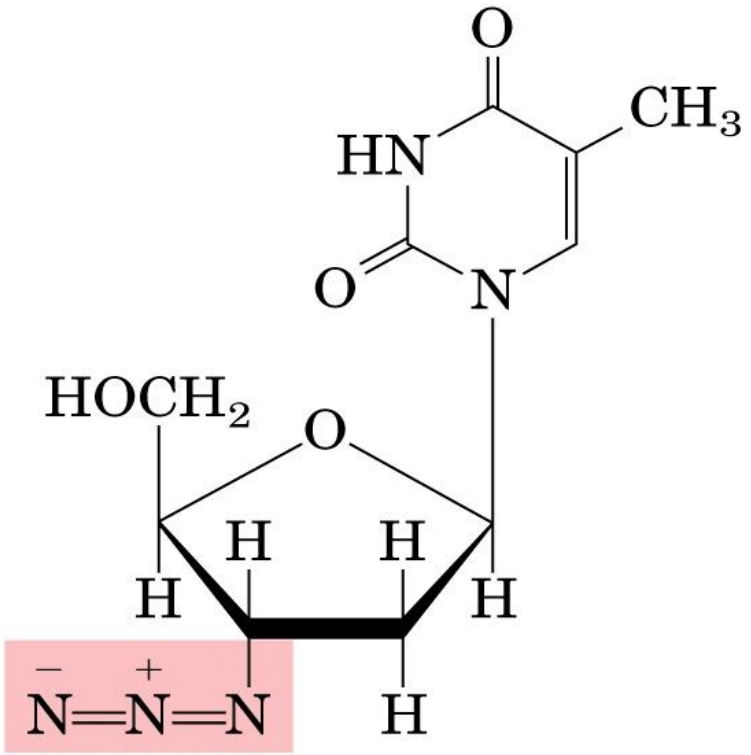
Inhibits HIV , telomerase and other reverse transcriptases.

Failed as an anticancer drug.

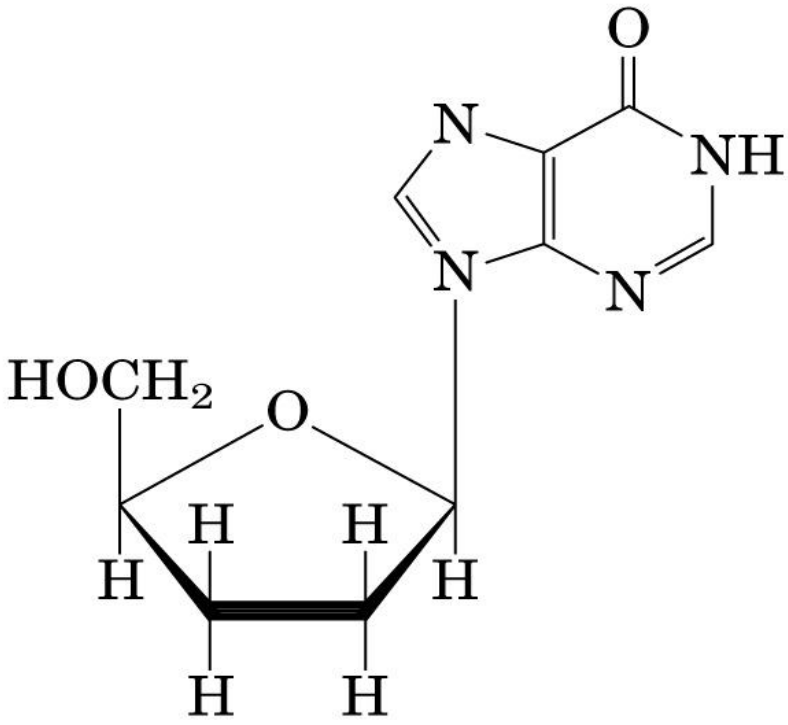
AZT (can enter plasma mb)→convertd to AZT triphosphates (cant)

HIV reverse transcriptase (not cellular/ host) has higher affinity to AZT than dTTP

→termination of viral DNA synthesis.



3′-Azido-2′,3′-dideoxythymidine (AZT)



2′,3′-Dideoxyinosine (DDI)

Retroviral genome:

Gag group associated antigen.

Pol protease

Env envelop

LTR sequences for initiation

& regulation of transcription.

Ψ viral packaging .

